

Adaptive Quality Mask Harvesting for Pixel-Wise Quality-Weighted Deep-Sky Reconstruction

Method specification, implementation constraints, and an M31-A run

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Abstract

Adaptive Quality Mask Harvesting (AQMH) is a pixel-wise quality-weighted reconstruction method for short-exposure deep-sky imaging. It replaces coarse tile-local quality scores with dense per-frame quality maps and reconstructs each supported output pixel by a deterministic weighted mean whose AQMH weight is evaluated at the same pixel. The method is independent of Classic Tile Compile: registration, normalization, common overlap, reporting, and run management may be shared, but AQMH does not use classic local tile metrics, state clustering, or synthetic frames as reconstruction inputs or fallbacks. This paper consolidates the AQMH v0.1.0 method, describes the implementation contract used by tile_compile, and evaluates a concrete M31-A run from 2026-06-11. The run processed 645 frames on a 3924×2310 canvas, wrote 645 dense quality maps at half resolution, and reconstructed the image with 5.699×10^9 finite map samples, zero missing map samples, zero unsupported pixels, and no fallback to classic tile weights. The measured median FWHM improved from 12.351 px to 5.271 px, corresponding to 57.327% improvement; background RMS changed from 4.185 to 0.714, a -82.932% change.

Keywords: astronomical image reconstruction, deep-sky imaging, quality maps, pixel-wise weighted mean, short exposure stacking, deterministic reconstruction

1 Introduction

Short-exposure deep-sky imaging collects many noisy frames instead of a small number of long exposures. The data contain useful signal even when individual frames suffer from local defects: cloud edges, satellite traces, registration remnants, small hot-pixel clusters, or spatially varying seeing. A single global frame score cannot model this spatial variation. Coarse tile scores improve locality, but they still assign one scalar quality value to an extended region. This produces two limitations: small artifacts can depress a large block, and constant weights inside adjacent blocks can introduce discontinuities.

AQMH is documented here as an independent reconstruction path within the tile_compile project. The reference method for Classic Tile Compile is [Tile-Based Quality Reconstruction for DSO, Version 3.3.9](#).

AQMH addresses this by computing a dense quality field $Q_{f,c}^{map}(x,y)$ for each frame f and channel or stream c . The reconstruction then uses the frame sample itself and the AQMH weight field at the same pixel:

$$R_c(p) = \frac{\sum_f W_{f,c}^{aqmh}(p) I_{f,c}(p)}{\sum_f W_{f,c}^{aqmh}(p)}, \quad W_{f,c}^{aqmh}(p) \geq 0. \quad (1)$$

This keeps the reconstruction core a deterministic

weighted mean after the weights have been computed. AQMH therefore does not hallucinate pixel values, does not synthesize replacement frames, and does not rely on tile clustering.

2 Method overview

2.1 Inputs and independence

AQMH reuses only shared preprocessing outputs:

- calibrated, registered, prewarped, and normalized frames;
- a canvas-valid/common-overlap mask;
- global frame diagnostics used for a frame-level weight;
- run management, artifacts, reports, and UI status.

It is otherwise independent from Classic Tile Compile. In particular, the run studied here reports `classic_tile_weights_used=false` and `fallback_to_classic=false`. The classic phases `STATE_CLUSTERING` and `SYNTHETIC_FRAMES` are not part of AQMH reconstruction.

2.2 Binding invariants

The AQMH implementation follows five practical invariants:

1. No frame is discarded solely because of its AQMH quality map.
2. Canvas-invalid pixels are excluded from all map and reconstruction statistics.
3. Once weights are known, reconstruction is a linear weighted mean of finite samples.
4. Quality maps and diagnostics are deterministic for identical input data and parameters.
5. AQMH outputs weights, masks, and diagnostics only; it does not predict missing signal.

The M31-A run uses `cherry_pick_enabled=false`; therefore no optional pixel-level frame subset mode is active.

2.3 Quality-map model

For a normalized registered input frame $I_{f,c}(x, y)$, AQMH computes a multi-scale quality field. At each scale s , three local signals are evaluated over valid canvas support:

- a sharpness signal based on local Laplacian-response variance;
- a local SNR signal based on background-centered positive signal over robust scale;
- an artifact anomaly gate based on high-pass outlier fraction.

The sharpness and SNR signals are normalized within a frame by robust z-scores and combined by a sigmoid:

$$\Psi_s(x, y) = \sigma(w_{sharp}z(\Phi_{sharp,s}) + w_{snr}z(\Phi_{snr,s})) \times \Phi_{artifact,s}(x, y). \quad (2)$$

Scale maps are upsampled with valid-support interpolation and fused by geometric mean:

$$Q_{f,c}^{map}(x, y) = \left(\prod_{s \in S_{actual}} \Psi_s^{up}(x, y) \right)^{1/|S_{actual}|}. \quad (3)$$

The geometric mean is intentionally conservative: one poor scale suppresses the fused quality at that location. Canvas-invalid pixels are set to zero only as an output convention after they have already been excluded from all statistics.

2.4 Reconstruction

The effective AQMH weight multiplies the frame-level quality term and the spatial quality map:

$$W_{f,c}^{aqmh}(x, y) = G_{f,c} Q_{f,c}^{map}(x, y). \quad (4)$$

For each output pixel, the implementation accumulates finite, canvas-valid samples only. Missing quality maps are not replaced by unweighted mean contributions. A zero quality map is an explicit veto for that frame/pixel contribution.

Table 1: M31-A run geometry and frame counts.

Quantity	Value
Input frames loaded	645
Usable frames	645
Canvas size	3924×2310 px
Common-overlap fraction	0.975
Reconstruction fraction	0.982
Adaptive tiles in shared grid	209
Tile size / stride	296 px / 208 px

Table 2: AQMH map storage and reconstruction support.

Quantity	Value
AQMH enabled	true
Frames/maps	645 / 645
Stored map size	1962×1155 px
Resolution divisor	2
Map dtype	float32
Finite samples	5.699×10^9
Missing samples	0
Unsupported output pixels	0
Zero-veto pixels	0
Classic fallback	false

3 Implementation run

3.1 Data and configuration

The evaluation data are taken from the `tile_compile` run on the analysis workstation:

```
tile_compile_cpp/runs/M31-A_20260611_193628
```

The paper bundle itself is archived under [docs/AQMH/zenodo-0.1.0](#) in the project repository. Run artifacts used for the measurements are local analysis outputs located in:

```
tile_compile_cpp/runs/M31-A_20260611_193628/
  artifacts
```

The final preview shown in Figure 4 is derived from `outputs/stacked_rgb_hms.fits`.

3.2 Quality-map diagnostics

The run wrote one luma quality map per frame. The average map mean was 0.496 with standard deviation 0.002. The per-frame map mean ranged from 0.488 to 0.500. The average artifact fraction was 0.051, with a range from 0.049 to 0.060. All 645 diagnostics reported the `scene_dependent_snr` flag, which is expected for the current AQMH SNR fallback path in this dataset and should be interpreted as a diagnostic marker rather than a reconstruction failure.

3.3 Validation metrics

The validation artifact reports an AQMH method run with both FWHM and background checks passing. The median input seeing estimate was 12.351 px, while the median output FWHM was 5.271 px. This corresponds to a 57.327% improvement. Background RMS changed from 4.185 to 0.714, reported as -82.932%.

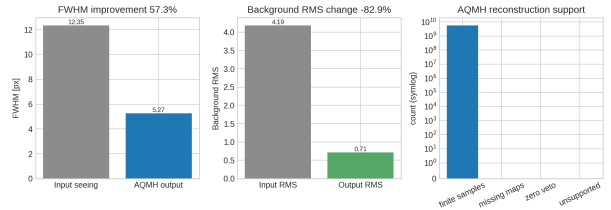
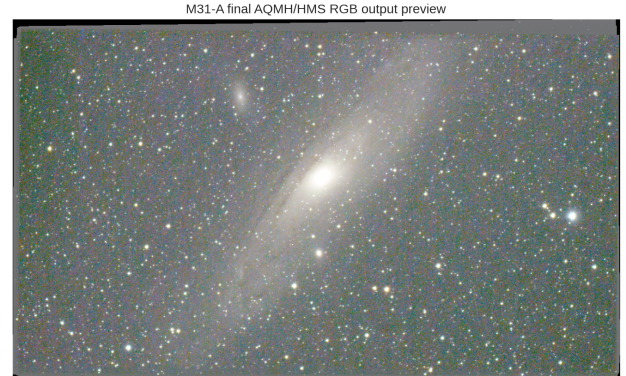


Figure 3: Validation summary for the M31-A AQMH run. FWHM and background RMS both improve, while the reconstruction support counters show no missing maps, unsupported pixels, or zero-veto pixels.



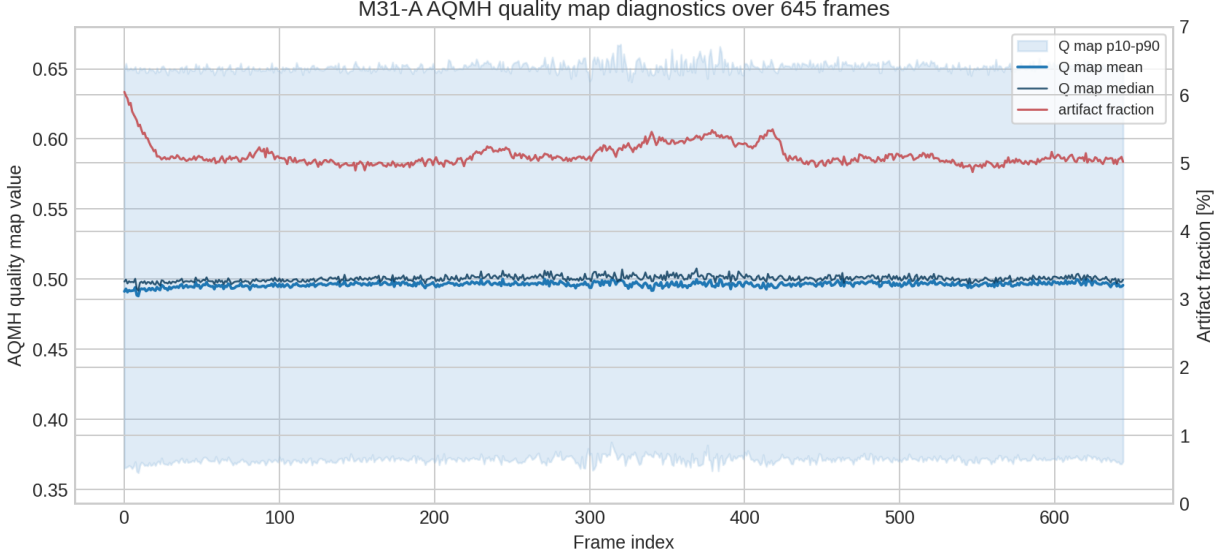


Figure 1: Per-frame AQMH quality-map diagnostics. The shaded band shows the p10-p90 interval of the map values; mean and median remain stable across the 645-frame sequence. The artifact fraction stays near five percent, with a higher early-frame tail.

Figure 4: Final RGB preview after AQMH reconstruction, PCC, and HMS. The preview is a visual run product derived from the FITS output, not a quantitative measurement.

- the measured FWHM and background RMS validation metrics improved.

This is important because AQMH is not a cosmetic replacement for the classic tile phases. It is a separate reconstruction core. The classic shared tile grid remains useful as an artifact and geometry diagnostic, but it is not the AQMH weighting model.

4 Discussion

4.1 What the run demonstrates

The M31-A run demonstrates the intended AQMH execution path:

- every usable frame produced a quality map;
- reconstruction consumed AQMH maps without classic tile-weight fallback;
- no quality map samples were missing during reconstruction;
- the output had full valid support under the run’s reconstruction mask;

4.2 Interpretation of stable map means

The frame-to-frame mean of Q^{map} is deliberately stable. The sharpness and SNR terms are z-normalized inside each frame, so their sigmoid contribution is mostly a within-frame relative field. Cross-frame quality discrimination comes from the global frame weight $G_{f,c}$ and from absolute veto terms such as the artifact gate. Therefore, a narrow distribution of map means is not a defect; it is a consequence of separating within-frame spatial quality from between-frame global quality.

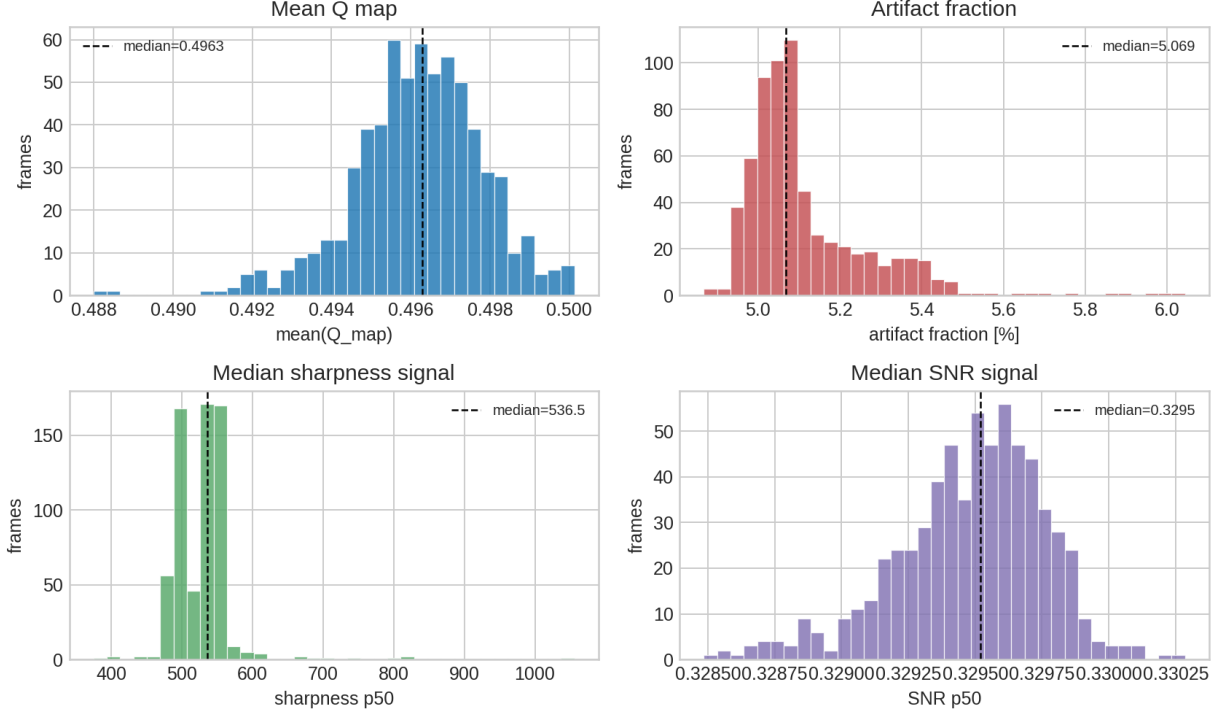


Figure 2: Distributions of AQMH per-frame diagnostic quantities. The narrow quality-map mean distribution reflects the within-frame relative normalization of the sigmoid factor; the artifact gate remains an absolute veto term.

4.3 Limitations

This paper reports one successful M31-A run and should not be read as a full statistical validation across object classes, sensors, and seeing regimes. The diagnostic flag `scene_dependent_snr=true` appears for all frames and deserves further evaluation: the fallback is deterministic, but future work should quantify when background-centered local SNR support is sufficient without the degraded-path flag. The final preview is also not a scientific photometry product; it includes downstream PCC and HMS processing.

5 Conclusion

AQMH provides a deterministic pixel-wise quality-weighted alternative to coarse local tile weighting for short-exposure deep-sky reconstruction. The M31-A run confirms that the implementation can execute the complete AQMH path on a 645-frame dataset, write all dense quality maps, reconstruct without classic fallback, and produce improved validation metrics. The observed FWHM improvement of 57.327%, the absence of missing map samples, and the stable quality-map diagnostics support the practical viability of AQMH as an independent reconstruction method in `tile_compile`.

Code and Data Availability

Reference method. Classic Tile Compile is described by [Tile-Based Quality Reconstruction for DSO, Version 3.3.9](#). The corresponding repository source is the [v3.3.9 paper source](#).

Code. The public project repository is github.com/jeamy/tile_compile. AQMH method details are documented in the [AQMH method document](#).

This paper. The AQMH v0.1.0 paper bundle is archived in [docs/AQMH/zenodo-0.1.0](#). The compact values used for the tables and plots are stored in [m31_aqmh_run_summary.json](#).

Run data. The full source measurement artifacts are local outputs from the M31-A analysis run and are not part of the public repository bundle.